Helius

Documentation Infrastructure Audit Report

Prepared by VeriOps

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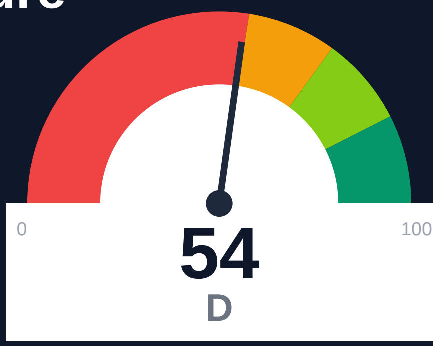


Risk: Critical

Key Metrics

Pages scanned: 328

Report type: Premium Executive Audit

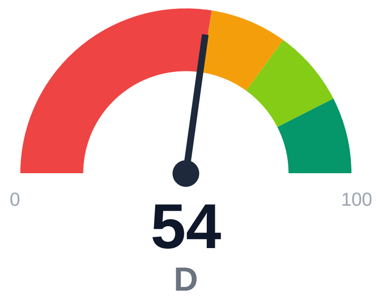
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 3
- API usage-doc coverage is low: 18.18% (260 API pages detected)
- Example reliability estimate is low: 0.0%

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Helius documentation audit covered 328 pages with 100% crawl coverage and 91.5% overall confidence. The site shows strong retrieval readiness (83.76 overall score, 65.24% chunkability) and complete actionability coverage. Critical gaps include 18.18% API reference coverage across 260 detected API pages, 0.0% example reliability, and 3 confirmed broken internal links. Evidence coverage stands at 83.33% (555 of 666 claims supported), with 15.55% of content flagged for staleness risk.

External posture: SEO/GEO issue rate **8.7%**, public API coverage **18.2%**, broken links **3**.

54 Audit Score	**F** Grade	**Critical** Risk Band
328 Pages Scanned	**3** Broken Links	**8.7%** SEO Issue Rate

Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://helius.dev/docs	328	3	18.2	0.0	8.7

Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	**92+**	**-38**
Industry average (developer docs)	**85**	**-31**
Minimum acceptable	**70**	**-16**
Your score	**54**	**-**

Gap to industry average: **31 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics

API Coverage (Public)

18.2%

SEO/GEO Issue Rate (lower is better)

8.7%

Example Reliability (estimate)

0.0%

Freshness Visibility (last-updated metadata)

100.0%

Internal Metrics (from scorecard)

API Coverage (Internal)

18.2%

Example Reliability (Internal)

0.0%

Docs-Contract Drift

16.7%

Pipeline Opportunity Fit (Deterministic)

This section maps cold-audit signals to problem classes that the pipeline can remediate directly. No LLM inference is used in this mapping.

LLM Retrieval Readiness (deterministic proxy)

83.8%

Audit Confidence (crawl scope + verification quality)

90.0%

Readiness band: **Strong**. Confidence: **High**. Open remediation opportunities: **4**.

SEO/GEO Quality Signal (inverse issue rate)

65.2%

Example Reliability Signal

0.0%

Freshness Visibility Signal

100.0%

API Coverage Signal

18.2%

LLM retrieval readiness breakdown (deterministic)

Chunkability

65.2%

Answerability

100.0%

Citationability

100.0%

Stale Risk (inverse)

84.5%

Evidence coverage: **83.3%** (555/666 key claims backed by deterministic evidence signals).

Highest-value remediation targets

Problem	Severity	Pipeline solution path
Internal documentation link integrity issues	MEDIUM	Weekly link governance via automated audits and consolidated remediation queue
Search and AI readability quality gaps	MEDIUM	Deterministic SEO/GEO linting and fix cycles before publish
Code example reliability risk	HIGH	Snippet lint + smoke checks + protocol self-verify coverage
API reference coverage and drift risk	HIGH	API-first contract flow, validators, regression gates, and drift checks

Financial Exposure Model

Low Estimate

$196,889
annual exposure

Remediation: **$1,615**
Monthly loss: **$3,188**
Opportunity: **$13,085**/mo

Base Estimate

$280,878
annual exposure

Remediation: **$4,208**
Monthly loss: **$9,970**
Opportunity: **$13,085**/mo

High Estimate

$421,680
annual exposure

Remediation: **$6,800**
Monthly loss: **$21,488**
Opportunity: **$13,085**/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-API-COVERAGE	Undocumented API operations	HIGH	$2,244	$850
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	HIGH	$1,734	$722
F-DRIFT	Code/docs contract drift	MEDIUM	$1,469	$510
F-FRESHNESS	Documentation freshness debt	LOW	$867	$425
F-LAYERS	Missing required doc layers	LOW	$1,377	$638
F-TERMINOLOGY	Terminology inconsistency	LOW	$536	$298
F-RETRIEVAL	RAG retrieval quality risk	LOW	$1,744	$765

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-API-COVERAGE	Undocumented API operations	HIGH
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	HIGH
F-DRIFT	Code/docs contract drift	MEDIUM
F-FRESHNESS	Documentation freshness debt	LOW
F-LAYERS	Missing required doc layers	LOW

Recommended Actions

1. Add usage examples and integration guides for the 213 API pages (81.82%) currently missing reference coverage [impact: critical, effort: high]

2. Implement automated example testing pipeline to validate code samples and lift 0.0% reliability estimate [impact: critical, effort: medium]

3. Fix 3 confirmed broken internal links to restore navigation integrity [impact: high, effort: low]

4. Audit and refresh the 15.55% of pages flagged for staleness risk to prevent outdated AI answers [impact: high, effort: medium]

5. Improve structural markup on the 34.76% of pages with low chunkability to optimize RAG retrieval [impact: medium, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

+ 100% crawl coverage (328 of 334 requested pages) with high confidence (91.5%)
+ Strong retrieval readiness band (83.76 score) with 100% answerability and citationability
+ Complete last-updated metadata coverage (100%) and full actionability coverage
+ 83.33% of key claims (555 of 666) include supporting evidence
+ Zero repository navigation broken links; all 3 broken links are user-facing docs

Risks

- 18.18% API reference coverage leaves 81.82% of 260 API pages without usage documentation, blocking developer onboarding
- 0.0% example reliability estimate suggests code samples may be outdated, untested, or non-executable
- 3 confirmed broken internal links degrade user trust and navigation flow
- 15.55% stale content risk may surface outdated information in AI-generated answers
- 65.24% chunkability indicates structural issues in 34.76% of pages that may hurt RAG retrieval quality
- 8.69% SEO/geo issue rate may limit discoverability in search engines
- 111 claims (16.67%) lack supporting evidence, reducing credibility for technical decision-makers

Limitations

* External audit cannot verify runtime correctness of code examples or confirm they execute against current API versions
* API coverage detection relies on heuristic page classification; actual endpoint-to-doc mapping requires internal API schema comparison
* Staleness risk flags are based on metadata and content patterns, not semantic accuracy or alignment with current product behavior
* Chunkability scoring measures structural markup quality but cannot assess conceptual clarity or completeness for human readers

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	85.0
support_hourly_usd	55.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	6.0
avg_release_delay_hours	18.0
monthly_support_tickets	280.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	25.0
monthly_doc_influenced_evals	45.0
eval_to_customer_rate	0.18
avg_customer_monthly_value_usd	1200.0
docs_friction_revenue_sensitivity	0.35

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	LOW

F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.